

Deep Learning for Optical Character Recognition for Non Latin-based Alphabets

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1 Introduction

Optical Character Recognition (OCR) is the process of converting text in images into machine readable text. This technology is extremely useful for digitizing texts, This task focuses on developing a Optical Character Recognition (OCR) system for non-Latin-based alphabets, which include scripts such as Greek, Cyrillic, Arabic, and Hebrew, to name a few. However, these alphabets have already been extensively studied, with multiple systems already in place to digitize these texts with a high degree of accuracy. Therefore, to ensure that this study explores an area with less prior research where resources are more limited and there is a higher lack of digital accessibility, the focus is directed to an alphabet which has had less technological advancement, Which left the Odia script as a perfect solution which is a classical and popular language in the Indian subcontinent used by more than 50 million people. In spite of its rich history, popularity and usefulness, not much research efforts have been made to achieve human level accuracy in case of Odia OCR.(Dash 2017).

1.1 Aims

The primary goal of this project is to develop software capable of accurately identifying characters from the Odia alphabet. This will rely on the use of neural networks, which are well-suited for learning complex patterns from data. Specifically, the use of a Convolutional Neural Network (CNN) will be utilized due to its effectiveness in recognizing patterns through multiple layers of processing. However, training such a model requires a substantial amount of data and due to the fact the Odia script has not been extensively studied and lacks large or many available datasets at all, an alternative approach is necessary. To solve this issue a script with similar structural attributes and features is identified to train the model. This approach is known as transfer learning and enables the model to use knowledge gained from the similar script to compensate for the limited data available for Odia.

1.2 Objectives

1. **Research Similar alphabets:** Research any scripts that are similar to Odia and have been studied in OCR systems, with large amounts of data available.
2. **Character set Research:** Study the similarities and differences between Odia and potential scripts to identify the best source script for transfer learning.
3. **Data Collection and Preprocessing:** Collect datasets for both Odia and the identified similar script. Apply preprocessing steps such as normalizing, and labeling images to ensure compatibility with the model.

4. **Base Model Development:** Implement a Convolutional Neural Network (CNN) for character recognition using the identified similar script's dataset.
5. **Transfer Learning:** Fine-tune the pre-trained model to recognize Odia characters through transfer learning.
6. **Evaluation:** optimize performance of the model on the Odia dataset and evaluate its accuracy.

2 Progress report

2.1 Research

To develop an effective OCR system for Odia characters the first step was to research to identify any suitable scripts for transfer learning. Bengali and Assamese scripts were closely examined as potential candidates due to the fact they both belong to the Indo-Aryan language family, However due to Bengali datasets being larger and more accessible compared to Assamese datasets along with the fact that both Bengali and Assamese scripts share some structural similarities with Odia, Bengali exhibits a higher degree of overlap, particularly in shared characters, stroke styles, and overall structural features. Another reason is that Bengali OCR systems have been more extensively studied, offering a wealth of information, pretrained models, and resources that can serve as a strong foundation for transfer learning such as Safrir 2021 which highlights the process of creating a OCR for Bangla script end to end. Assamese OCR research, on the other hand, is relatively limited and was much harder to find many similar studies.

2.2 Data Collection and Pre-processing

The next step was to locate a well-structured dataset of handwritten and printed Bengali character set to be the primary source for transfer learning. For this part the Banglalekha Isolated dataset was used which is designed for Bengali handwritten character recognition. It consists of 84 character classes, including 50 basic characters, 10 numerals, and 24 compound characters. Also containing around 166,000 images of handwritten characters (Banglalekha 2023) making it the perfect choice.

Along with that a suitable Odia Dataset was required the one used in this project is the Odia Characters Dataset By Saswat Samal 2017 which consists of around 300 images each pre-labelled and organised all being 50 x 50 greyscale.

After collection the next step was preprocessing which included multiple steps starting with python using it to standardize and prepare the data for training the CNN. All images were resized to a uniform size of 50x50 pixels, ensuring consistent dimensions for all inputs in the model. Additionally, all images were converted to grayscale, reducing

complexity while preserving the essential features necessary for character recognition using libraries such as OpenCV and NumPy. These libraries were utilized to complete tasks like resizing and greyscale conversion efficiently.

To ensure the Odia dataset aligns with the Bengali dataset, similar preprocessing steps were applied. These included labeling to ensure all character images were correctly categorized into distinct classes, resizing all images to 50x50 pixels, and converting them to greyscale. enabling the model to effectively leverage features learned from Bengali characters and adapt them to the recognition of Odia script. This was done to ensure that transfer learning would be able to work effectively as the inputs would be in the same format.

2.3 Optimization

Research confirmed that transfer learning is a viable approach for this project. The shared structural features between Bengali and Odia scripts, such as circular strokes, vertical symmetry, and similar diacritical marks, allow the features learned from Bengali data to be effectively transferred to Odia.

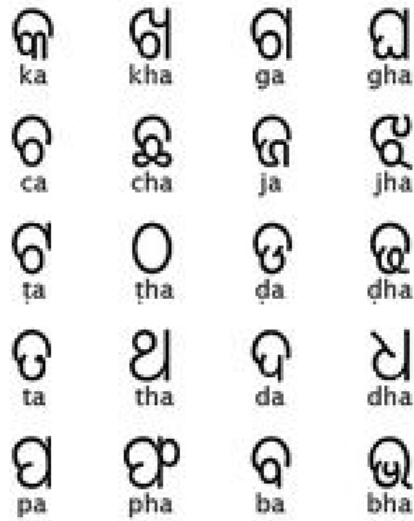


Figure 1: Odia characters and their circular nature

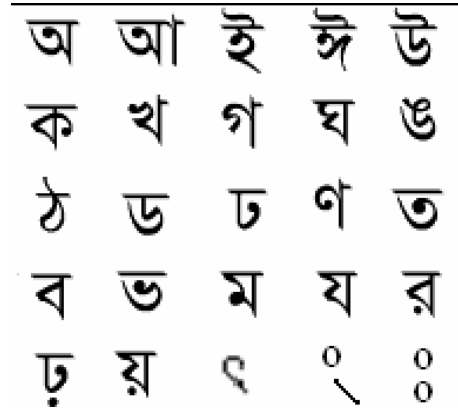


Figure 2: Bengali characters with similar structural features

This approach eliminates the need to train the model from scratch on the Odia dataset and works since its relatively smaller and less diverse as you can see in Figure 1. By leveraging transfer learning, the project can save time and computational resources while achieving better accuracy.

3 Progress Evaluation and Risk Analysis

3.1 Problems Encountered

The following problems that were faced during the development process:

1. **Limited Availability of Odia Data:** While Bengali datasets are easy to access, the availability of high-quality and labelled Odia datasets is significantly harder to find. The process of collecting a usable Odia dataset has been very time-consuming and presented challenges such as:
 - Finding data that is unlabelled.
 - Computer-generated text rather than being handwritten samples.
 - Insufficient amounts of data.
2. **Preprocessing Challenges:** Ensuring compatibility between the Bengali and Odia datasets required extensive preprocessing, including resizing, normalization, and grayscale conversion. Challenges faced during this stage include:
 - Inconsistent image dimensions and quality in the raw data, requiring significant intervention.
 - Ensuring labeling and class organization for both datasets to avoid errors during training which was very time consuming throughout.
3. **Risk of Overfitting:** Due to the relatively small size of the Odia dataset compared to the Bengali dataset, there is a risk of the model overfitting to the Odia training data, which could impact its ability to generalize to new samples.
4. **Time Constraints:** The collection and preprocessing phases have taken longer than expected, leaving less time for model development, training, and evaluation. However the project is still on track with only a week delay to start the coding phase as seen in Figure 3.

3.2 Future Problems

Insufficient Model Accuracy on Odia Data Despite leveraging transfer learning, the structural differences between Bengali and Odia scripts could lead to suboptimal model performance on the Odia dataset. The model may struggle with recognizing unique or rarely occurring Odia characters that lack direct counterparts in the Bengali script.

Dataset Bias Both Bengali and Odia datasets may introduce bias due to uneven representation of certain characters, writing styles.

3.3 Addressing Problems

Expand Odia Dataset Collection to address overfitting risk using techniques such as data augmentation techniques like rotation or scaling which means rotating the data by a certain angle to get a different piece of data or scaling data up or down to create your own data. Also using regularization techniques such as dropout and weight decay during model training could also help stop overfitting and ensure the model is actually learning. Another technique that could be used is cross-validation which can be used to evaluate the model's ability to generalize to unseen data.

To address the time constraint caused by the two-week delay in the design, the next step is to prioritize tasks that directly affect model performance, like training the base model, and defer less critical objectives to future iterations if needed.

3.4 Conclusion

A lot of progress has been made in the planning for the development of the OCR system. Through extensive research and analysis, Bengali was identified as a suitable script for transfer learning, and dataset collection and preprocessing have been completed. Whilst also facing challenges such as limited Odia data availability and preprocessing complexities have been encountered and a clear plan of action has been created to address these issues. The next steps will be more focused on implementing the neural network, applying transfer learning, and fine-tuning the model to achieve optimal performance. With these measures in place and a clear structure, The project remains on track to meet its objectives.

4 Appendices

References

- [1] K. S. Dash, N. B. Puan, and G. Panda. *Odia character recognition: a directional review*. Artificial Intelligence Review, 48(4): 473–497, 2017. <https://doi.org/10.1007/s10462-016-9507-5>.
- [2] F. B. Safir, A. Q. Ohi, M. F. Mridha, M. M. Monowar, and M. A. Hamid. *End-to-End Optical Character Recognition for Bengali Handwritten Words*. 2021 National Computing Colleges Conference (NCCC), Taif, Saudi Arabia, 2021, pp. 1–7. <https://doi.org/10.1109/NCCC49330.2021.9428809>.
- [3] Shipa. *BanglaLekha Isolated Train Test Split*. Kaggle. <https://www.kaggle.com/datasets/shipa129/banglalekha-isolated-train-test-split>.

- [4] S. Samal. *Odia Characters Dataset*. Kaggle. <https://www.kaggle.com/datasets/saswatsamal/odia-characters-dataset>.

